

## Supplier or Vendor Rating Definition

**Vendor rating** is when the suppliers are provided with a status or a title based on several factors.

Vendors can earn a rating based on how reliable, punctual, and reasonably priced their goods are — not to mention the quality, of course!

Ratings range from Good or Average all the way up to Best-in-Class — depending on performance and your company's overall criteria for success.

## Vendor Rating Criteria



When it comes to rating suppliers, there are key elements that buyers take into consideration of the following things.

### Quality

The quality of the products or goods the vendor supplies is the main factor. The vendor can maintain good quality by improving production and having quality planning in the [supply chain](#). Quality factor consists following things.

- The supplier should follow the terms and conditions mentioned in the purchase order.
- The vendor's products or services must meet the specifications mentioned in the proposal and purchase order request.
- The product failure rate should be within the appropriate limit.

- The vendor should do proper repair or rework.
- He should provide an adequate time duration for replacement.

### **Price**

A company always wants to get the materials at less expense to reduce its manufacturing cost to increase its profit. Hence the vendor needs to set a competitive price for his products. It includes the following things.

- **Stable price:** The price of the product or service must be stable over time.
- **Accurate price:** There should not be much difference between the purchase order and invoice prices.
- **Prior notice about price changes:** He should inform about price changes in advance.
- **Billing:** He must provide easily readable and understandable bills.

### **Delivery**

The supplier has to develop the ability to deliver the goods on a scheduled date. This factor consists following things.

- **Lead time:** Lead time is between the actual delivery day and order placement day. The shortest lead time helps to get a good impression of the supplier. The vendor should deliver products on or before the promised date.
- **Quantity:** He must deliver the correct amount of products as mentioned in the contract.
- **Packing and documentation:** Packing of the products must be suitable, studied, and undamaged. The vendor should provide proper documents along with the delivered products.
- **Emergency delivery:** The vendor must be able to deliver products in case of emergency requirements.

### **Service**

It is one of the crucial criteria for the supplier. He has to provide good service by providing an updated catalog, pricing, technical information, etc.

- The vendor must have the ability to handle complaints effectively.
- The vendor should provide technical support for installation, maintenance, and repair.
- Emergency support: The supplier should support in the emergency condition of product failure or repair.
- Resolve the problems: The supplier should find the solution for the problem on time.

### **10 C's of Supplier Evaluation**

10C's of supplier evaluation is defined by Dr. Carter, According to Carter, an organization must consider these 10 C's while rating or evaluating its vendors.

They are:

- **Competency** — How competitive is the supplier?
- **Capacity** — How capable is the supplier of fulfilling requirements?
- **Commitment** — Does the vendor committed to supplying expected quality products or services?
- **Control** — Does the supplier have control over policies and procedures?
- **Cash** — What is the financial status of the supplier?
- **Cost** -What is the cost offered by the supplier for his products? Is it worth it?
- **Consistency** — Does the supplier able to provide quality products or services consistently?
- **Culture** — What is the business culture of the supplier? Is that culture match your culture?
- **Clean** — Does the vendor follow sustainable practices and green management rules? Does his business ethical?
- Communication** -What is the communication process? What is the response time?

### **How do you Improve your Vendor's Performance?**

Here is a list of tips to encourage and improve supplier performance ratings.

Make higher management participate in the process

It is essential because you can only get sufficient resources to execute this process if you involve higher management.

Set a performance measurement

Set key performance indicators (KPIs) and make sure that they cover areas like vendor service, vendor response capacity, vendors' managing skills, and sales support. Let them know that you are monitoring them and reviewing their performance.

Clearly define your expectations

Communicate with the vendors about what you want and what they need to focus on. For example, provide a unique requirements list to each vendor instead of giving all vendors a general requirements list so they can quickly identify your needs.

Do a realistic analysis of the supply chain management system

Explore your purchase and supply chain management practices to discover their strengths and weaknesses.

Have a good team for supplier management

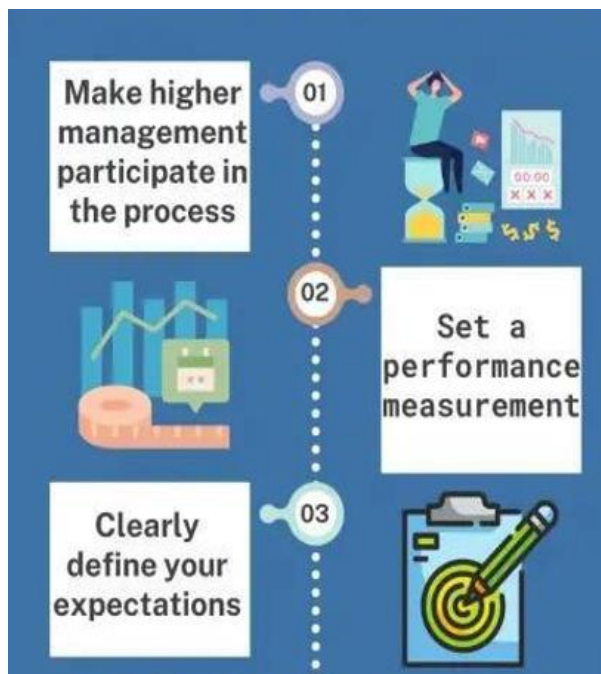
Based on the complexity of the supply chain, have team members for quality checks, logistics, planning, and engineering. In addition, let the team members share supplier management plans and targets so that this team approach helps to boost vendor performance.

Train your team members

Train them about approaching the vendors, making them understand your needs, and reaching their goals. While training, concentrate on what you want from your vendors and how to approach them.

Keep a good relationship with your vendor

Analyzing and evaluating the vendors is insufficient. Instead, have a friendly relationship with them. Visit their place, and greet them when they come to your place. It improves their confidence and performance.



**What Are the Steps for Evaluating Supplier Performance?**

Evaluating supplier performance involves more than an assessment for onboarding. According to Keeley the following steps are :

1. Start evaluating supplier performance regularly using formal and sophisticated assessment. “It’s never too late to start,” Keeley said.
2. Establish supplier performance evaluation criteria and take steps to install standards, policies, processes and procedures.
3. Develop a supplier performance evaluation form. This will standardize the assessment and make decisions better, faster and more strategic.
4. Establish, qualify and classify suppliers based on supplier development aims.
5. Review data and reach out to suppliers again. Usually, the more critical the supplier is to your business, the more often you need to touch base.
6. Use digital technology to connect the data and make best use of the information you have.
7. Hold conversations with suppliers to develop plans on how to improve any areas that are suboptimal.
8. Put actions into place that will correct these areas.
9. Follow through strategically. Step back and re-evaluate supplier performance based on your criteria.
10. Repeat the process.

This questionnaire developed such as it explore the following criteria / attributes:

- 1) Quality of product
- 2) Past performance/experience
- 3) Financial situation of supplier
- 4) Human resource availability
- 5) Equipment capabilities
- 6) Price
- 7) Capacity to supply
- 8) Maintaining relations

- 9) Location of site
- 10) Delivery at a time

### Supplier's rating

Here's a simple example to calculate a supplier's rating based on the criteria provided in the document:

#### Example Scenario:

A supplier delivers production materials, and their performance is evaluated based on **Quality**, **Supply Performance**, and **Purchasing**. The total score is calculated using the weighted percentages: **Quality (45%)**, **Supply Performance (40%)**, and **Purchasing (15%)**.

#### Supplier Data:

##### 1. Quality:

- **PPM Value:** 150 ppm → 60 points
- **Claim Rate:** 1.5% → 60 points
- **Audit Result:** Overall B rating → 60 points
- **Claim Management:** Good → 80 points
- **Weighted Average for Quality:** (PPM: 20%, Claim Rate: 20%, Audit Result: 5%, Claim Management: 45%) =  $(60 \times 0.2) + (60 \times 0.2) + (60 \times 0.05) + (80 \times 0.45) = \mathbf{66 \text{ points}}$

##### 2. Supply Performance:

- **Adherence to Delivery Dates:** Delivery 2 days late → 60 points
- **Additional Freight Costs:** 1 additional trip → 80 points
- **Weighted Average for Supply Performance:** (Adherence: 35%, Freight Costs: 5%) =  $(60 \times 0.35) + (80 \times 0.05) = \mathbf{62 \text{ points}}$

##### 3. Purchasing:

- **Competitiveness:** Good → 80 points
- **Quality Certificate:** ISO 9001 → 60 points

- **Environmental Certificate:** Environmental questionnaire answered → 60 points
- **Cooperation:** Satisfactory → 60 points
- **Weighted Average for Purchasing:** (Competitiveness: 3%, Quality Certificate: 3%, Environmental Certificate: 3%, Cooperation: 6%)  

$$= (80 \times 0.03) + (60 \times 0.03) + (60 \times 0.03) + (60 \times 0.06) = \mathbf{63 \text{ points}}$$

#### **Total Score Calculation:**

- **Quality (45%):**  $66 \times 0.45 = \mathbf{29.7}$
- **Supply Performance (40%):**  $62 \times 0.40 = \mathbf{24.8}$
- **Purchasing (15%):**  $63 \times 0.15 = \mathbf{9.45}$

**Total Score** =  $29.7 + 24.8 + 9.45 = \mathbf{63.95 \text{ points}}$

#### **Classification:**

Based on the total score:

- **63.95 points** falls under **C classification (< 70 points)**.
- The supplier is classified as a **C supplier**, indicating serious deficits and immediate blocking for inquiries and new orders.

**90% - 100% (Top Tier):** Preferred partner status and more business.

**70% - 89% (Acceptable):** Standard ongoing business relationship.

**Below 70% (Poor):** Triggers a mandatory performance fix or vendor replacement.

#### **What factors influence the claim rate?**

The claim rate is influenced by several factors, including:

1. **Quality of Delivered Goods:** The physical quality of the materials or products supplied directly impacts the number of claims. Poor-quality goods lead to higher claim rates.
2. **Supplier Processes:** Inefficient or inconsistent manufacturing and quality control processes at the supplier's end can result in defective goods, increasing the claim rate.
3. **Communication and Specifications:** Miscommunication or unclear specifications between the supplier and the buyer can lead to mismatched expectations, resulting in claims.
4. **Complexity of Products:** More complex products or components are more prone to defects, which can increase the claim rate.
5. **Volume of Deliveries:** A higher volume of goods delivered increases the likelihood of defects, potentially raising the claim rate.
6. **Inspection and Testing:** The thoroughness of inspection and testing processes at both the supplier's and buyer's ends can influence the claim rate. Poor inspection may allow defects to go unnoticed initially.
7. **Supplier's Quality Management System:** Suppliers with robust quality management systems (e.g., certified to IATF 16949 or ISO 9001) are likely to have lower claim rates due to better control over their processes.
8. **Environmental or External Factors:** Issues such as transportation damage, storage conditions, or external disruptions can also contribute to defects and claims.

By addressing these factors, suppliers can work to reduce their claim rate and improve their overall performance.

## **Vendor Rating Example**



Let us rate two companies, A and B.

Factors considered are quality, price, and delivery.

Weights for each of the above factors. That is

- Quality – 60%
- Price – 20%
- Delivery – 20%

If we multiply each factor's values by their weights, we can derive ratings and compare which is better.

#### Company A inputs:

**Total quantity supplied:** 10 units, total quantity accepted: 8 units, price per unit: \$10, Delay in delivery 20% time delay.

$$\text{Quality rating} = (8/10) \times 100 = 80\%$$

$$\text{Price rating} = (\text{Price Ratio Lowest} / \text{Simpler Price}) \times 100$$

$$= (10/10) \times 100 = 100\%$$

$$\text{Delivery rating} = 100 - 20 = 80\%$$

$$\text{weighted vendor rating of company A} = (80 \times 60\% + 100 \times 20\% + 80 \times 20\%) = 84$$

#### Company B inputs:

Total quantity supplied: 20 units, total quantity accepted: 18 units, price per unit: \$16, Delay in delivery 10% delay.

$$\text{Quality rating} = (18/20) \times 100 = 90\%$$

$$\text{Price rating} = (\text{Price Ratio Lowest} / \text{Simpler Price}) \times 100$$

$$= (10/16) \times 100 = 62.5\%$$

$$\text{Delivery rating} = 100 - 10 = 90\%$$

$$\text{weighted supplier rating of company B} = (90 \times 60 + 62.5 \times 20 + 90 \times 20) = 84.5$$

### **Company C inputs:**

Total quantity supplied: 25 units, total quantity accepted: 20 units, price per unit: \$15, Delay in delivery 15% delay.

$$\text{Quality rating} = (20/25) \times 100 = 80\%$$

$$\text{Price rating} = (\text{Price Ratio Lowest} / \text{Simpler Price}) \times 100$$

$$= (10/15) \times 100 = 66.66\%$$

$$\text{Delivery rating} = 100 - 15 = 85\%$$

$$\text{weighted supplier rating of company C} = (80 \times 60\% + 66.66 \times 20\% + 85 \times 20\%) = 48 + 13.33 + 17$$

$$= 78.33$$

Though the price per unit of Company B is more than Company A and C, Company B still wins because of its overall rating.

### **Supply performance**

*Adherence to delivery dates and quantities, additional freight costs (SF)*

The responsible MRP controller rates the adherence to schedules and quantities by comparing the delivery quantity and delivery date with target specifications.

Should delivery generally take place on the agreed date (1 day earlier max.), we award 100 points. Otherwise the following deductions apply:

Deliveries generally too early

Delivery 1 day early = 100 points

Delivery 2 days early = 80 points

Delivery 3 days early = 60 points

Delivery 4 days early = 40 points

Delivery > 4 days early = 1 point

### **Some deliveries delayed**

Delivery 1 day late = 80 points

Delivery 2 days late = 60 points

Delivery 3 days late = 40 points

Delivery > 3 days late = 1 point

### ***Additional freight costs (additional trips) (SF)***

No additional trips = 100 points

1 additional trip caused = 80 points

2 additional trips = 60 points

3 additional trips = 40 points

> 3 additional trips = 1 point

Knockout criterion with respect to soft facts: If the supplier provides for a special customer status, all soft facts are rated as 1, regardless of other criteria.

### ***Purchasing***

#### ***Competitiveness (price/performance ratio) (SF)***

Very good = 100 points

Good = 80 points

Satisfactory = 60 points

Poor = 40 points

Insufficient = 1 point

### ***Quality certificate***

International Automotive Task Force (IATF) 16949 = 100 points

German automotive industry (Verband der Automobilindustrie) VDA 6.1 = 80 points

DIN ISO 9001 = 60 points

No certificate = 1 point

### ***Environment***

DIN EN ISO 14001 = 100 points

Environmental questionnaire answered = 60 points

Environmental questionnaire not answered = 1 point

### ***Cooperation (SF)***

Very good = 100 points

Good = 80 points

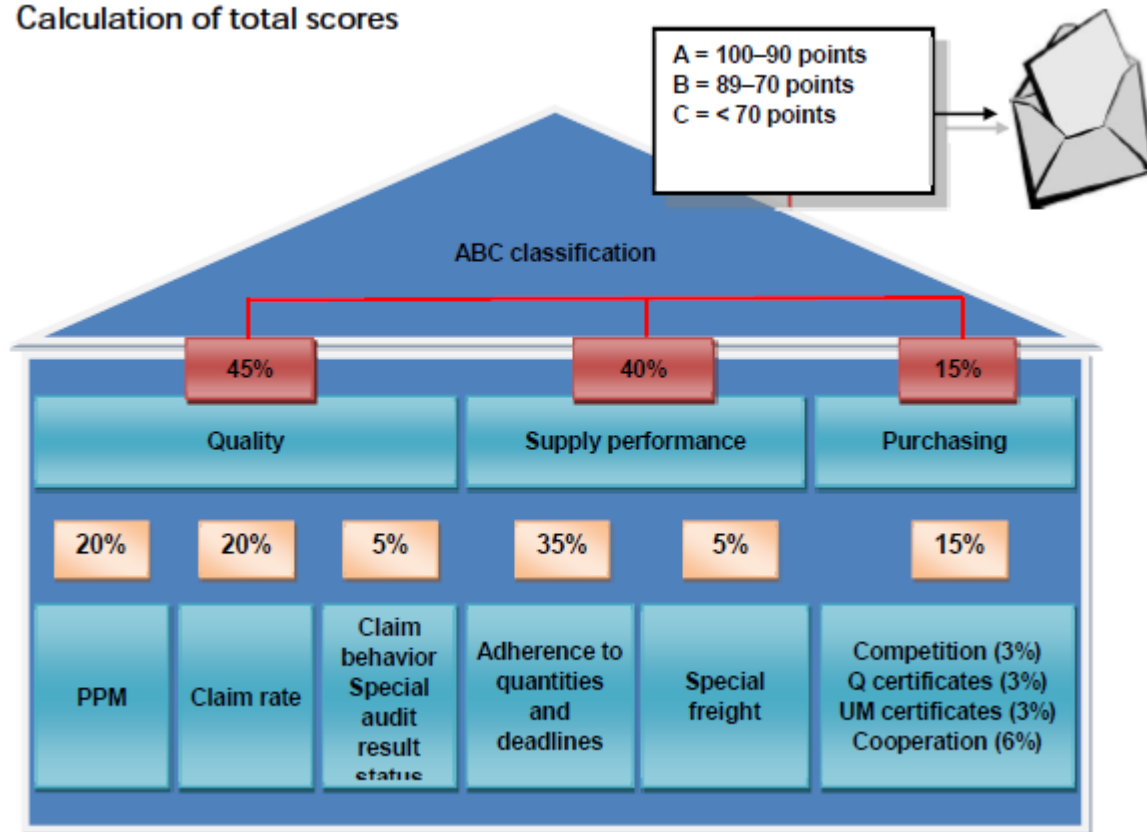
Satisfactory = 60 points

Poor = 40 points

Insufficient = 1 point

Calculation of total scores

## Calculation of total scores



## Classification

### A suppliers

A suppliers enjoy preferred status in the event of new projects and orders, and deductions do not apply when bids are compared. A suppliers have proven themselves as reliable partners and work to maintain and improve high levels of performance.

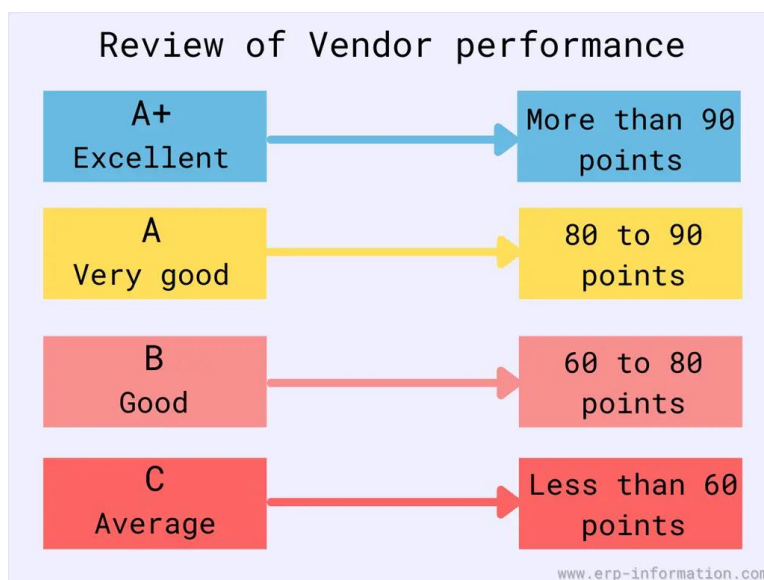
### B suppliers

B suppliers are those who we deem to have potential for improvement. Deductions apply to B suppliers when bids are compared. In the event of price equality in offers to an A-supplier, this A-supplier shall be preferred when placing the order. To discuss possible improvements with you, you fall under escalation level E1 (see Escalation Process).

### C suppliers

C suppliers there are serious deficits between our requirements and those of our customers with regard to the services you provide. A successful cooperation is not given, therefore you are blocked immediately for inquiries and new orders. You receive the special status E2 according to our escalation process.

### Vendor Rating Example



Let us rate two companies, A and B.

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Weights for each of the above factors is

- Quality — 60%

- Price — 20%

- Delivery — 20%

If we multiply each factor's values by their weights, we can derive ratings and compare which is better.

**Company A inputs:**

Total quantity supplied: 10 units, total quantity accepted: 8 units, price per unit: \$10, Delay in delivery 20% time delay.

Quality rating =  $(8/10) \times 100 = 80\%$

Price rating =  $(10/10) \times 100 = 100\%$

Delivery rating =  $100 - 20 = 80\%$

**weighted vendor rating of company A** =  $(80 \times 60 + 100 \times 20 + 80 \times 20) = 84$

### **Company B inputs:**

Total quantity supplied: 20 units, total quantity accepted: 18 units, price per unit: \$16, Delay in delivery 10% delay.

Quality rating =  $(18/20) \times 100 = 90\%$

Price rating =  $(10/16) \times 100 = 62.5\%$

Delivery rating =  $100 - 10 = 90\%$

**weighted supplier rating of company B** =  $(90 \times 60 + 62.5 \times 20 + 90 \times 20) = 84.5$

Though the price per unit of company B is more than company A, company B still wins because of overall rating is high.

### **Vendor Rating**

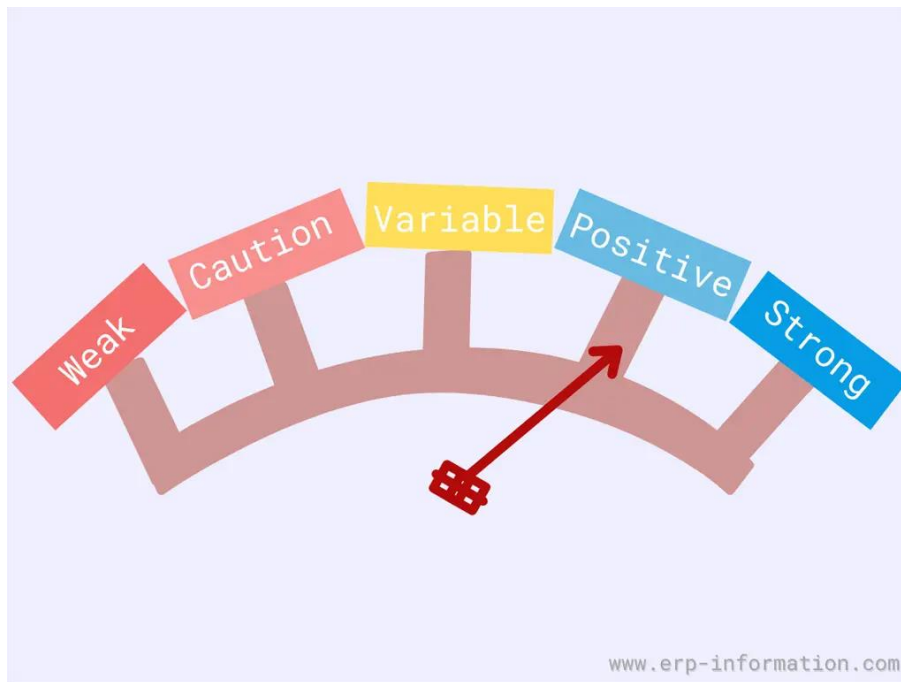
Gartner vendor rating is a method that provides the best content to the customer to evaluate vendors or providers for mutual benefits.

It considers things like,

- Product/Service offered by the vendor
- Support provided by the vendor
- The pricing structure of the product or services
- Technology
- Strategy

- Corporate Viability

It rates the vendor from weak to strong.



- **Strong:** Vendors with the best focus, quality products or services, and a good market position will come under this category. The customer feels comfortable and shows interest in continuing the relationship with the vendor. He considers this provider as the best choice for new investments.

- **Positive:** Vendors with good focus and quality products/services will come under the positive category. Customers can continue working with vendors for planning investments.

- **Variable:** Average vendors will come under this category. The customer should consider the maturity of the vendor and the short-term & long-term effects of the relationship.

- **Caution:** Poor service providers will come under this type.

The [customer](#) should understand the limitations of the vendor and eventually

plans based on risk and future business effects.

- **Weak:** Vendors with inferior performance will come under this category. Customers should consider this vendor only for planned investment and eventually arrange another vendor.

## Example

A supplier rating can be calculated using weighted averages of performance metrics like quality and delivery. For example, one formula is  $Sr = 0.6(Qr) + 0.4(Dr)$ , where  $Sr$  is the supplier rating,  $Qr$  is the quality rating, and  $Dr$  is the delivery rating. The quality rating is often calculated based on the proportion of accepted items out of the total quantity supplied ( $Qr = \frac{Qa}{Qs}$ ), while the delivery rating can factor in on-time delivery (

$$Dr = \frac{Qty.receivedintime + Qty.delayed(1 - K * Td/T)}{TotalQty.ordered}).$$

## Example calculation

**Scenario:** You place a purchase order for 100 units from a supplier. The supplier is supposed to deliver within 5 days.

### 1. Calculate Quality Rating ( $Qr$ )

- **Quantity Accepted ( $Qa$ ):** 98 units
- **Quantity Rejected ( $Qrr$ ):** 2 units
- **Total Quantity Supplied ( $Qs$ ):**  $Qa + Qrr = 98 + 2 = 100$  units
- $Qr = \frac{Qa}{Qs} = \frac{98}{100} = 0.98$

## 2. Calculate Delivery Rating ( $Dr$ )

- Quantity ordered ( $Q$ ): 100 units
- Promised delivery time ( $T$ ): 5 days
- Quantity supplied in time ( $Qt$ ): 95 units
- Quantity delayed ( $Qd$ ): 5 units (shipped one day late)
- Time delay for quantity delayed ( $Td$ ): 1 day
- Constant ( $K$ ): 2

$$• Dr = \frac{Qt + Qd(1 - K \times \frac{Td}{T})}{Q} = \frac{95 + 5(1 - 2 \times \frac{1}{5})}{100} = \frac{95 + 5(1 - 0.4)}{100} = \frac{95 + 5(0.6)}{100}$$

$$= 95 + 3/100 = 98/100 = 0.98$$

## 3. Calculate Overall Supplier Rating ( $Sr$ )

- $Sr = 0.6(Qr) + 0.4(Dr)$
- $Sr = 0.6(0.98) + 0.4(0.98) = 0.588 + 0.392 = 0.98$  

To calculate a supplier's rating, define key performance indicators (KPIs) like quality and delivery, assign a weight to each, and calculate a weighted average score using your data. For example, a simple rating could be calculated as: Overall Score = (Quality Score  $\times$  0.4) + (Delivery Score  $\times$  0.3) + (Cost Score  $\times$  0.2) + (Compliance Score  $\times$  0.1). You will need to collect consistent data on each supplier's performance for each metric to perform the calculation.

### 1. Identify KPIs and assign weights

- **Identify key performance indicators:** Common metrics include quality, on-time delivery, cost, and responsiveness.
- **Assign weights:** Decide on the importance of each KPI for your business needs. For instance, quality might be 40%, delivery 30%, cost 20%, and compliance 10%.

### 2. Calculate scores for each KPI

- **Quality:** Can be calculated based on the number of accepted lots versus total lots received, or by using metrics like parts per million (ppm) defects.
- **Example (Accepted Lots):** Quality Score = (Quantity Accepted / Quantity Received) x 100
- **Example (PPM):** PPM Score = (Defective Parts / Delivered Quantity) x 1,000,000. You can then convert this ppm value to a score based on your own scale.

Defect rate

$$\frac{\text{Total Number of Defective Units}}{\text{Total Number of Units Inspected}} \times 100$$

example, if a supplier delivers 5,000 units of a part, and 200 of those are defective, the defect rate is 4%.

- **Delivery:** Often calculated as a fill rate or on-time delivery percentage.
- **Example (Fill Rate):** Delivery Rating = (Units Received / Units Ordered) x 100
- **Example (On-Time):** On-Time Score = (Number of on-time deliveries / Total deliveries) x 100
- **Cost and Compliance:** Gather data from financial records and compliance documentation to score these areas.
- **Service/Responsiveness:** Use a subjective scale (e.g., 1-5) based on factors like communication and support response times.

### 3. Calculate the overall weighted score

- Multiply the score for each KPI by its assigned weight.
- Add the results together to get the final overall score.

- **Example: Overall Score = (Quality Score × 0.4) + (Delivery Score × 0.3) + (Cost Score × 0.2) + (Compliance Score × 0.1).**

#### **4. Analyze and interpret the results**

- Use the final score to compare suppliers and identify top performers or those needing improvement.
- Look for trends over time to understand long-term performance.

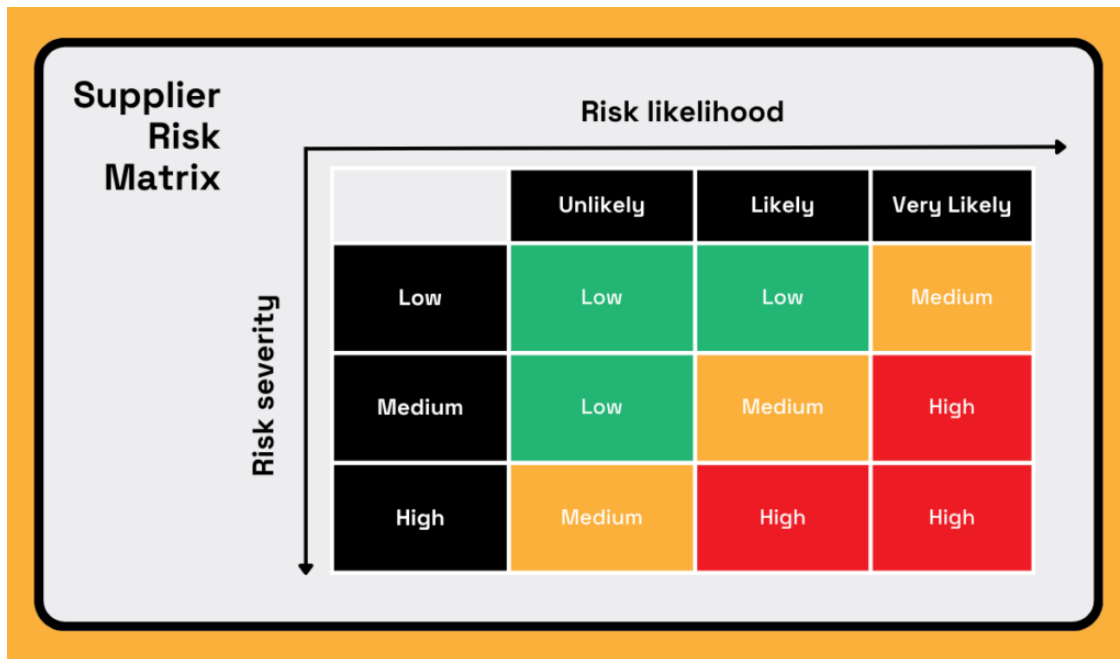
#### **Return Rate**

Similarly, the return rate is a key measure of product quality.

It is the percentage of products returned due to defects or other issues.

This is calculated by dividing the number of returned units by the total units delivered and multiplying by 100.

$$\frac{\text{Number of Returned Units}}{\text{Total Units Delivered}} \times 100 = \text{Return Rate}$$



Tools like supplier scorecards, [the Kraljic Matrix](#), and other risk matrices, which you can see above, are useful for visualizing these risks and deciding how critical a supplier is to your operations.